

Data Analysis Lab for Machine Learning Introduction

Objective:

The goal of this lab is to introduce basic data analysis techniques using a sample dataset. You will learn how to:

1. Load and explore data.
 2. Use simple visualisations like histograms, line graphs, and bar charts to gain insights from the data.
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Step 1: Setup

Required Libraries:

We will use the following Python libraries for this lab:

- **pandas**: For data manipulation.
- **matplotlib**: For creating visualisations.
- **seaborn**: For more advanced and aesthetically pleasing graphs.

```
# Install the required libraries if needed
# !pip install pandas matplotlib seaborn

# Import the libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Set the style for seaborn plots
sns.set(style='whitegrid')
```

Step 2: Load the Dataset

For this lab, we will use a simple dataset, **student_scores.csv**, which contains data about students' study hours and their corresponding scores.

Dataset: student_scores.csv

Loading the Dataset:

```
# Load the dataset
data = pd.read_csv('student_scores.csv')

# Show the first few rows of the dataset
print(data.head())
```

Step 3: Explore the Dataset

Before diving into visualisations, let's do some basic exploration of the dataset.

1. **Display basic statistics** of the data such as mean, median, and standard deviation.

```
# Get basic statistics of the dataset
print(data.describe())
```

Check for missing values:

```
# Check for any missing values in the dataset
print(data.isnull().sum())
```

Step 4: Data Visualisation

1. **Histogram:** Visualise the distribution of Study Hours.

A histogram helps us see the distribution of continuous data, which is useful to understand how students' study hours are spread.

```
# Plot a histogram for 'Study Hours'
plt.figure(figsize=(8,6))
plt.hist(data['Study Hours'], bins=5, color='skyblue', edgecolor='black')
plt.title('Distribution of Study Hours')
plt.xlabel('Study Hours')
plt.ylabel('Frequency')
plt.show()
```

2. **Bar Chart:** Compare the test scores for different students.

A bar chart allows us to see how each student performed in the test.

```
# Plot a bar chart for 'Test Score'
plt.figure(figsize=(8,6))
plt.bar(data['Student'], data['Test Score'], color='orange',
edgecolor='black')
plt.title('Test Scores of Students')
plt.xlabel('Student')
plt.ylabel('Test Score')
plt.show()
```

3. **Line Plot:** Analyse the relationship between Study Hours and Test Score.

A line plot is useful to show trends and relationships between two continuous variables.

```
# Plot a line graph showing Study Hours vs Test Score
plt.figure(figsize=(8,6))
plt.plot(data['Study Hours'], data['Test Score'], color='green',
marker='o')
plt.title('Study Hours vs Test Score')
plt.xlabel('Study Hours')
plt.ylabel('Test Score')
plt.grid(True)
plt.show()
```

4. Scatter Plot: Display the correlation between Study Hours and Test Score.

A scatter plot helps to see if there's a linear relationship between two continuous variables.

```
# Scatter plot to show correlation between Study Hours and Test Score
plt.figure(figsize=(8,6))
sns.scatterplot(x='Study Hours', y='Test Score', data=data, color='red',
s=100)
plt.title('Study Hours vs Test Score')
plt.xlabel('Study Hours')
plt.ylabel('Test Score')
plt.show()
```

Step 5: Interpretation of the Results

1. **Histogram:** The histogram shows how many students fall into different ranges of study hours. If most of the data is skewed towards higher hours, it might indicate that student's study longer to score higher.
2. **Bar Chart:** This shows how individual students performed. From the bar chart, we can easily compare test scores between students.
3. **Line Plot:** The line plot illustrates how test scores change as study hours increase. If the line is mostly upward sloping, this suggests that more study hours lead to higher scores.
4. **Scatter Plot:** The scatter plot can help reveal any correlation between study hours and test scores. A clear upward trend in the points indicates a positive correlation, i.e., as study hours increase, test scores increase.

Step 6: Conclusion

This simple lab helped you learn how to:

1. Load and explore a basic dataset.
2. Create different types of visualisations like histograms, bar charts, line graphs, and scatter plots.
3. Draw conclusions from visual patterns.